EES 301: Global biogeochemical cycles and Climate change

Ocean chemistry

Table 10-7

Mean oceanic concentrations

Atomic number	Element	Species	Mean water concentration
1	Hydrogen	H_2O	
2	Helium	He (gas)	1.9 nmol/kg
3	Lithium	Li^+	178 μg/kg
4	Beryllium	$BeOH^+$	0.2 ng/kg
5	Boron	$B(OH)_3$	4.4 mg/kg
		$B(OH)_4^-$	
6	Carbon	ΣCO_2	2200 μmol/kg
7	Nitrogen	N_2, NO_2^-	590 μmol/kg
		NO_3, NH_4^+	30 μmol/kg
8	Oxygen	Dissolved O_2	150 μmol/kg
9	Fluorine	F^-, MgF^+	1.3 mg/kg
10	Neon	Ne (gas)	8 nmol/kg
11	Sodium	Na^+	10.781 g/kg
12	Magnesium	Mg^{2+}	1.28 g/kg
13	Aluminum	$Al(OH)_4^-$	1 μg/kg
14	Silicon	Silicate $Si(OH)_4$	110 μmol/kg
15	Phosphorus	Reactive phosphate	2 μmol/kg
16	Sulfur	Sulfate SO_4^{2-}	2.712 g/kg
17	Chlorine	Chloride Cl^-	19.353 g/kg
18	Argon	Ar (gas)	15.6 μmol/kg
19	Potassium	K^+	399 mg/kg
20	Calcium	Ca^{2+}	415 mg/kg (C.A.)
21	Scandium	$Sc(OH)_3^0$	<1 ng/kg
22	Titanium	$Ti(OH)_4^0$	<1 ng/kg
28	Vanadium	$H_2VO_4^-$, HVO_4^{2-}	<1 μg/kg
24	Chromium	Cr(tot)	330 ng/kg (Si)
		$Cr(OH)_3(s)$	330 ng/kg (Si, PO)
		CrO_4^{2-}	350 mg/kg (Si, NO)

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25	Manganese	Mn^{3+}, $MnCl^{+}$	10 ng/kg
26	Iron	$Fe(OH)_2^{+}$, $Fe(OH)_4^{-}$	40 ng/kg
27	Cobalt	Co^{2+}	2 ng/kg
28	Nickel	Ni^{2+}	480 ng/kg
29	Copper	$CuCO_3^{0}$, $CuOH^{+}$	120 ng/kg
30	Zinc	$ZnOH^{+}$, Zn^{2+}, ZnCO	390 ng/kg
31	Gallium	$Ga(OH)_4^{-}$	10–20 ng/kg
32	Germanium	$Ge(OH)_4^{0}$	5 ng/kg
33	Arsenic	$HAsO_4^{2-}$, $H_2AsO_4^{-}$	2 μg/kg
		Dimethylarsenate	
34	Selenium	[Se(tot)]	170 ng/kg
		SeO_3^{2-}	
		[SeIV]	—
		[SeVI]	—
35	Bromine	Br^{-}	67 mg/kg
37	Rubidium	Rb^{+}	124 μg/kg
38	Strontium	Sr^{2+}	7.8 mg/kg (PO_4)
			7.7 mg/kg (Sr/Cl)
39	Yttrium	$Y(OH)_3^{0}$	13 ng/kg
40	Zirconium	$Zr(OH)_4^{0}$	<1 μg/kg
41	Niobium	—	1 ng/kg
42	Molybdenum	MoO_4^{2-}	11 μg/kg
44	Ruthenium	—	0.5 ng/kg
45	Rhodium	—	—
46	Palladium	—	—
47	Silver	$AgCl_2^{-}$	3 ng/kg
48	Cadmium	$CdCl_2^{0}$	70 ng/kg
49	Indium	$In(OH)_2^{+}$	0.2 ng/kg
50	Tin	$SnO(OH)_3^{-}$	0.5 ng/kg
51	Antimony	$Sb(OH)_6^{-}$	0.2 μg/kg
52	Tellurium	$HTeO_3^{-}$	—
53	Iodine	IO_3^{-}, I^{-}	59 μg/kg (PO_4 corr.)
			60 μg/kg (NO_3 corr.)

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54	Xenon	Xe(gas)	0.5 nmol/kg
55	Cesium	Cs^{+}	0.3 ng/kg
56	Barium	Ba^{2+}	11.7 μg/kg
57	Lanthanum	$La(OH)_3^0$	4 ng/kg
58	Cerium	$Ce(OH)_3$	4 ng/kg
59	Praseodymium	$Pr(OH)_3$	0.6 ng/kg
60	Neodymium	$Nd(OH)_3$	4 ng/kg
61	Promethium	$Pm(OH)_3$	—
62	Samarium	$Sm(OH)_3$	0.6 ng/kg
63	Europeum	$Eu(OH)_3$	0.1 ng/kg
64	Gadolinium	$Gd(OH)_3$	0.8 ng/kg
65	Terbium	$Tb(OH)_3$	0.1 ng/kg
66	Dysprosium	$Dy(OH)_3$	1 ng/kg
67	Holmium	$Ho(OH)_3$	0.2 ng/kg
68	Erbium	$Er(OH)_3$	0.9 ng/kg
69	Thulium	$Tm(OH)_3$	0.2 ng/kg
70	Ytterbium	$Yb(OH)_3$	0.9 ng/kg
71	Lutetium	$Lu(OH)_3^0$	0.2 ng/kg
72	Hafnium	—	<8 ng/kg
73	Tantalum	—	<2.5 ng/kg
74	Tungsten	WO_4^{2-}	<1 ng/kg
75	Rhenium	ReO_4^-	4 ng/kg
76	Osmium	—	—
77	Iridium	—	—
78	Platinum	—	—
79	Gold	$AuCl_2^-$	11 ng/kg
80	Mercury	$HgCl_4^{2-}$, $HgCl_3^0$	6 ng/kg
81	Thallium	Tl^+	12 ng/kg
82	Lead	$PbCO_3^0$	1 ng/kg
83	Bismuth	BiO^+, $Bi(OH)_2^+$	10 ng/kg
84	Polonium	PoO_3^3, $PoO(OH)_2^0$	—
86	Radon	Rn (gas)	—
88	Radium	Ra^{2+}	—
89	Actinium	—	—
90	Thorium	$Th(OH)_4^0$	<0.7 ng/kg
91	Protactinium	—	—
92	Uranium	$UO_2(CO_3)_2^{4-}$	3.2 μg/kg

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Table 10-9

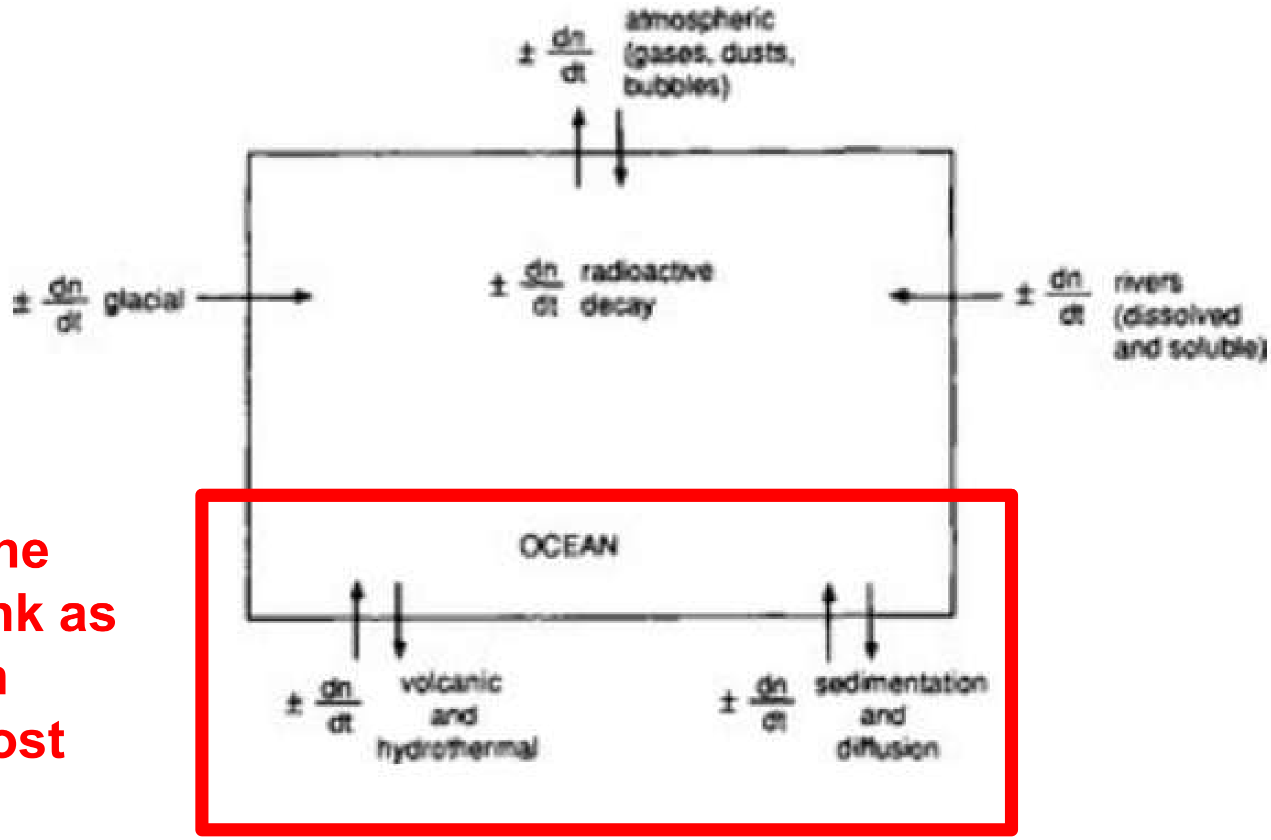
The major ions of seawater: concentration at 35‰, ratio to chlorinity, and molar concentration[a]

	The major ions of seawater		
Ion	g/kg at $S = 35‰$	g/kg (chlorinity ‰)	mol/L
Cl^-	19.354	0.9989	5.46×10^{-1}
SO_4^{2-}	2.712	0.1400	2.82×10^{-2}
Br^-	0.0673	0.00347	8.4×10^{-4}
F^-	0.0013	0.000067	6.8×10^{-5}
B	0.0045	0.000232	4.1×10^{-4}
Na^+	10.77	0.5560	4.68×10^{-1}
Mg^{2+}	1.290	0.0665[b]	5.32×10^{-3}
Ca^{2+}	0.4121	0.02127	1.02×10^{-2}
K^+	0.399	0.0206	1.02×10^{-2}
Sr^{2+}	0.0079	0.00041	9.1×10^{-5}
HCO_3^-	0.142	—	2.387×10^{-3}

[a]From Wilson (1975).

[b]Recent reported values lie between 0.06612 and 0.06692.

Residence time



Seems to be controlled by the lithospheric sink as this is the main sink flux for most elements

FIG. 10-17 A simplified box model of the ocean.

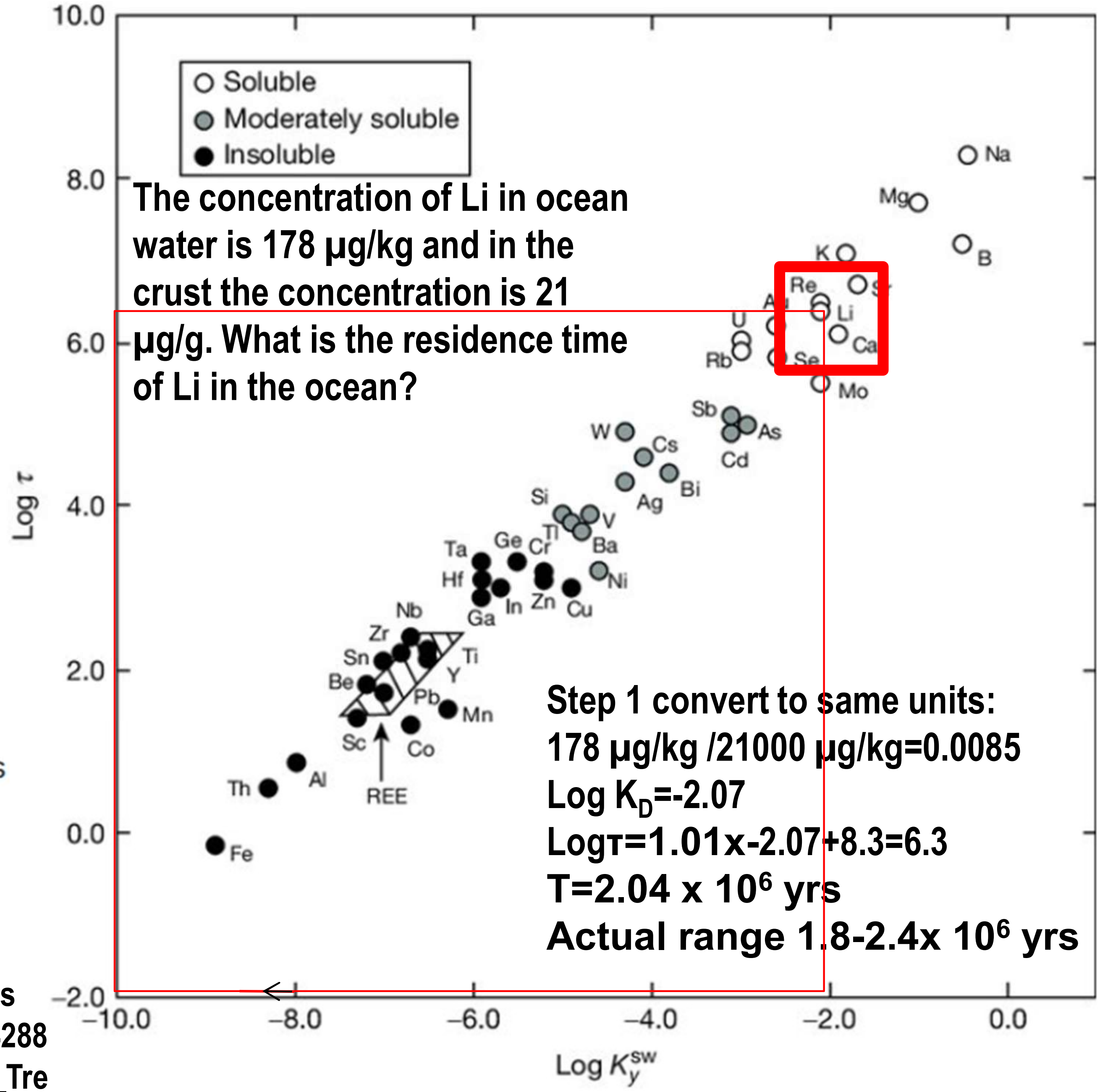
$$\tau_0 = \frac{\text{Mass of element in the sea}}{\text{Mass supplied (or removed) per year}} = \boxed{\frac{M}{S}} = \frac{M}{Q}$$

Residence time

$$\log \tau_0 = a \log K_D + b$$

a~1.01; b~8.4

For many elements there is a relationship between their lifetime and the ratio of their concentration in the sea water /crust that may reflect long term chemical controls



The concentration of Li in ocean water is 178 µg/kg and in the crust the concentration is 21 µg/g. What is the residence time of Li in the ocean?

Step 1 convert to same units:
178 µg/kg /21000 µg/kg=0.0085
Log K_D=-2.07
Logτ=1.01x-2.07+8.3=6.3
T=2.04 x 10^6 yrs
Actual range 1.8-2.4x 10^6 yrs

Figure 4 Plot of residence time (expressed as log t) against seawater upper crust partition coefficient (expressed as K sw Y) (source Taylor and McLennan, 1985).

Crustal composition can be found here, sea water one is in the book chapter/lecture slides https://www.researchgate.net/publication/234288836_Composition_of_the_Continental_Crust_Treatise_Geochem_31-64

What depth profiles tell about sources and sinks

Table 10-8

Characteristic types of profiles of elements in the ocean with example elements and probable controlling mechanisms

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca^{2+} + 2HCO_3^-$$

Since the solubility of CO_2 in water increases as temperature drops the exoskeletons of plankton will dissolve below a certain depth. For aragonite the dissolution depth is 1-2 km for calcite it is 4-5 km.

Type of profile	Example elements	Mechanisms
1. Similar to salinity	Na, K, Mg, SO_4, F, Br	Conservative elements of very low reactivity
2. Sea-surface enrichment		Atmospheric input
	^{210}Pb, Mn	– natural
	^{90}Sr, Pb	– pollution, bomb tests
	NO	– photochemistry
	$As(CH_3)_2$, H_2, NO_2	Biological production
3. Photic-zone depletion with deep-ocean enrichment	Ca, Si, ΣCO_2, NO_3, PO_4, Cu, Ni	Biological uptake and regeneration
4. Mid-water maxima		
3000 m	^{3}He, Mn, CH_4, ^{222}Rn	Hydrothermal input
200 to 1000 m	^{3}H	Isopycnal transport
	Mn, NO_2	Redox chemistry in oxygen minimum
5. Bottom-water enrichment	^{222}Rn, ^{228}Ra, Mn, Si	Flux out of the sediments
6. Deep-ocean depletion	^{210}Pb, ^{230}Th,Cu	Scavenging by settling particles

Air-sea Exchange

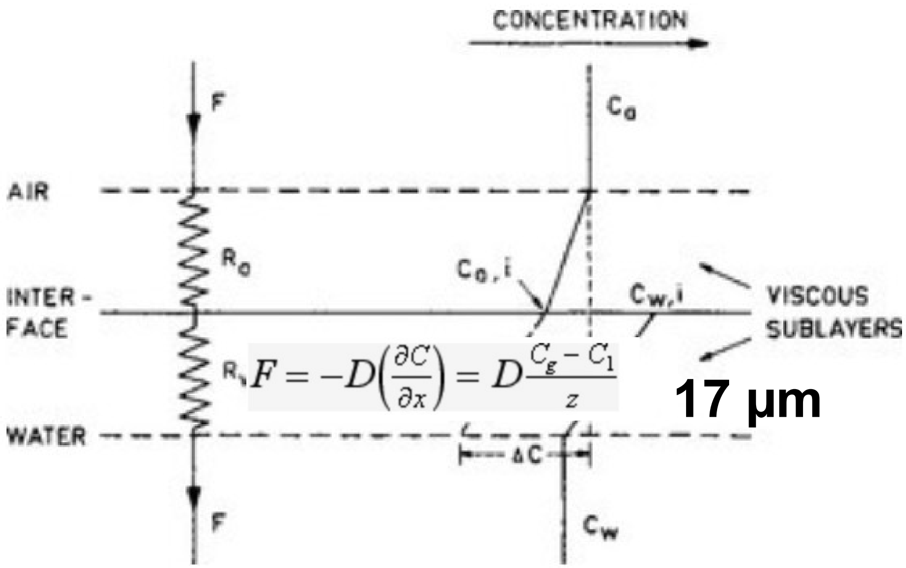


FIG. 4-16 A simplified model of flux resistances and concentration gradients in the viscous sublayers at the air-sea interface.

The flux is defined to be positive if directed from the atmosphere to the ocean and negative in the opposite direction.

It is proportional to the concentration gradient

$$F = k\Delta c$$

Which is dependent on the concentration on the atmosphere c_a and water c_w and the Henry's law constant K_H

$$\Delta c = c_a - K_H c_w$$

The parameter k that is being multiplied with Δc is the transfer velocity.

The reciprocal of the transfer velocity corresponds to the resistance to transfer across the surface. The total resistance ($R=k^{-1}$) can be viewed as the sum of the air resistance R_a and water resistance R_w.

$$R = R_a + R_w = \frac{1}{k_a} + \frac{K_H}{\alpha k_1}$$

In this the parameters k_a and k_1 are the transfer velocities for chemically unreactive gases through the viscous sublayers in the air an water respectively. The parameter α quantifies enhancement in the value of $k_1$due to chemical reactivity of the gas in the water.

Air-sea Exchange

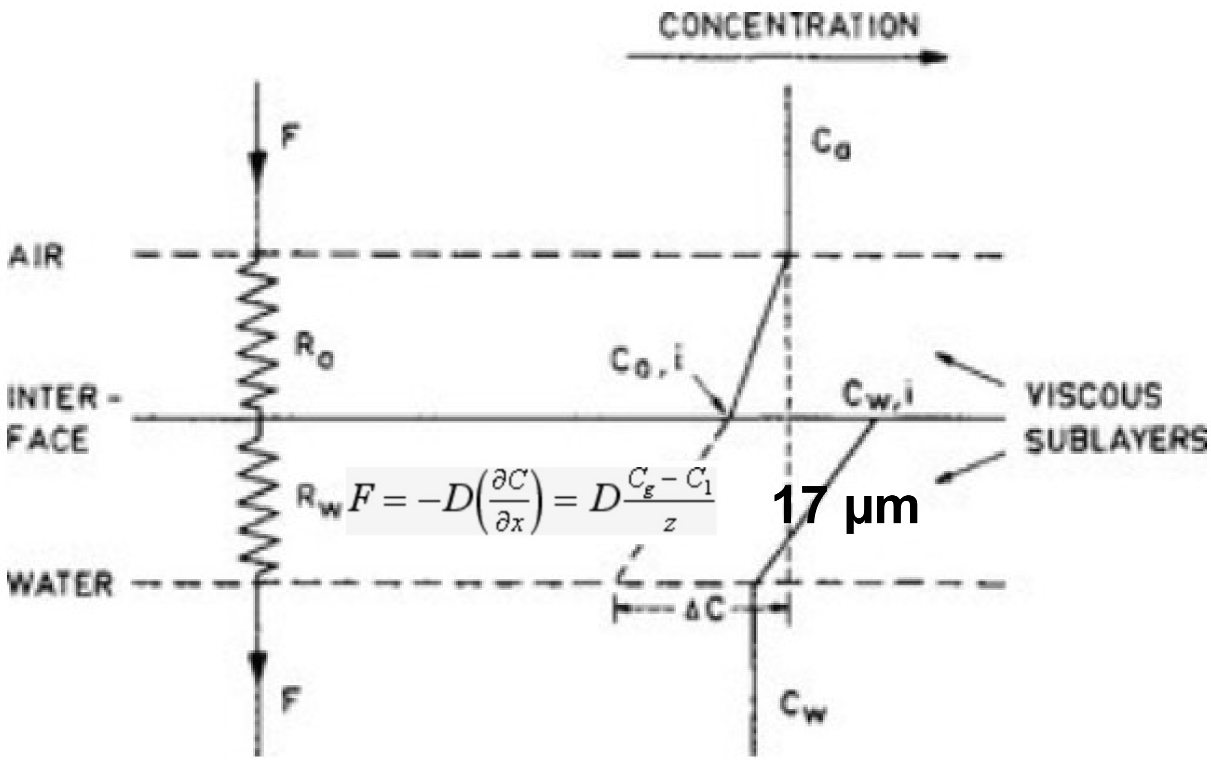


FIG. 4-16 A simplified model of flux resistances and concentration gradients in the viscous sublayers at the air-sea interface.

$T_{turb} = L^2/D$

When in the surface layer of the ocean mixed by wind (top 50 m) transport is dominated by turbulent diffusivity = 0.1 m^2/s

It takes:

$50^2/ 0.1$ s~7 h for the layer to exchange air

When laminar diffusion across the 17μm stagnant film is considered then D=1x10^{-5} cm^2/s it takes $0.0017^2/(1x10^{-5})$ =0.29 s to cross that layer 0.0017cm/0.29s so the ventilation becomes 5m/day or 1800 m/yr

Air-sea Exchange

Solubilities of various gases in surface ocean water

$$\tau_{atm} = \frac{\text{Mass in atmosphere}}{\text{Flux into the ocean}} = \frac{M_{atm}}{F}$$

$=M_{atm}/(K_H(D/z))f$

Gas	Partial pressure in dry air (atm)	Equilibrium concentration in surface seawater (cm^3/L) 0°C	24°C
H_2	5×10^{-7}	—	—
He	5.2×10^{-6}	4.1×10^{-5}	3.4×10^{-5}
Ne	1.8×10^{-5}	1.7×10^{-4}	1.5×10^{-4}
N_2	0.781	14	9
O	0.209	8.8	5.5
Ar	9.3×10^{-3}	0.36	0.22
CO_2	3.2×10^{-4}	0.47	0.23
Kr	1.1×10^{-6}	8.1×10^{-5}	4.9×10^{-5}
Xe	8.6×10^{-8}	1.2×10^{-5}	0.6×10^{-5}

$K_{H\,N2}$ =1.6 mol $m^{-3}atm^{-1}$
D/z= 0.0017cm/0.29s =5m/day or 1800 m/yr
M_{atm} surface pressure 1 atm = 101325 Pa; g=9.8 ms^{-2}
1 Pa=kg m^{-1} s^{-2}

M_{atm}=101325 [kg m^{-1} s^{-2}]/9.8 [m s^{-2}]=10339 [kg m^{-2}]=3.6 10^5 mol m^{-3}
Mass of air 29 g/mol

$K_{H\,N2}$ =1.6 mol $m^{-3}atm^{-1}$ 0.781 atm =1.2 mol m^{-3}
D/z=1800 m/yr

=3.6 10^5[mol m^{-2}]/(1.3×1800) [mol m^{-3}m yr^{-1}]
=158 yrs

Air-sea Exchange of particles from the ocean to the air

Particles are transported from the ocean to the air through bubble bursting by breaking waves called “whitecaps”. This process is wind speed dependent.

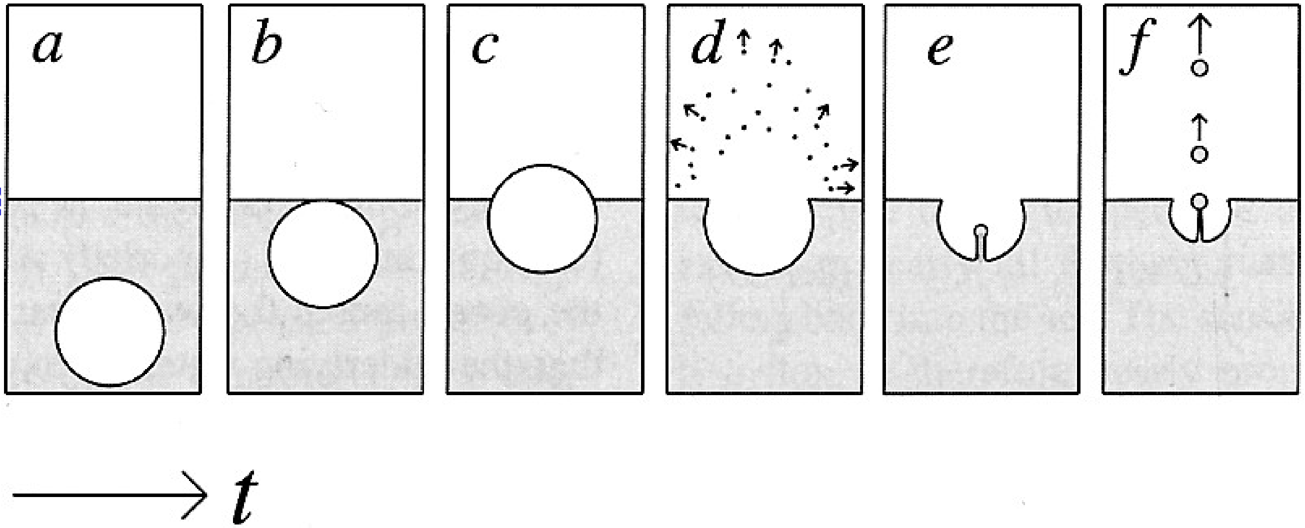
Seasalt aerosols...

wind→ bubbles→ spray

whitecap coverage W α U^{3+}!

seasalt production via bubble bursting...

- film drops (many, small, drags along surface organics)
- jet drops (fewer, larger, bulk)
- spume drops (larger still)



Air-sea Exchange of particles from the air to the ocean

Particles are transported via dry deposition F_d and wet deposition F_w after incorporation into rain drops. The fall velocity of both is driven by gravity. The usually larger rain drops fall much faster than smaller particles.

$$F = F_d + F_w = (\nu_w + \nu_d)c_a$$

The gravitational fall velocity depends on the mass of the particle determined by density ρ and diameter d, the viscosity η of air a shape correction factor χ for non-spherical shapes and a slip correction factor that is large for very small particles <100 nm.

$\eta_{air}=1.8 \times 10^{-5}$ Pa·s

$$\nu_{grav} = \frac{\rho_p d^2 C_c}{18\eta\chi_p}$$

Use this equation to calculate how far from the coast 5 µm sized hematite (5.3 g/cm^3) will iron fertilize the Atlantic ocean if the dust plume is originally lifted to 2 km and the trade winds blow with 15 m/s

When particles (e.g. dead plankton) fall in a dense fluid (ocean) you need to consider the density difference between the fluid and the particle, a step we neglect in the case of air.

$\eta_{ocean}=1.41 \times 10^{-3}$ Pa·s
For 35g/kg 10°C

$$\nu_{grav} = \frac{(\rho_p - \rho_f)d^2}{18\eta\chi_p}$$

Use this equation to calculate how long 0.5 mm foraminifera (2 g/cm^3) take to settle from 50 m depth to the ocean floor at 4000 m if the density of ocean water is (1.03 g/cm^3)

Sediment water exchange

Growth of sediments results from sedimentation and accumulation of solid particles and inclusion of water in the pore space between particles. The main processes are:

1. Sedimentation of solids (mineral, skeletal and organic materials)
2. Flux of dissolved material and water in to the sediment contribution to the growth of the sediment column
3. Upward flow of pore water and dissolved material caused by pressure gradients
4. Molecular diffusional fluxes in pore water.
5. Mixing of sediment and water at the interface through bioturbation and water turbulence

The rate of accumulation can vary from a few mm in 1000 years in the pelargic ocean up to centimeters per year in lakes and coastal areas.

The flux density of sediments is in the range of 0.006 to 6 kg/m^2 per year.

The accompanying flux density of materials dissolved in the trapped water is 10^{-6} to 10^{-3} kg/m^2 per year.

Biological processes in the ocean

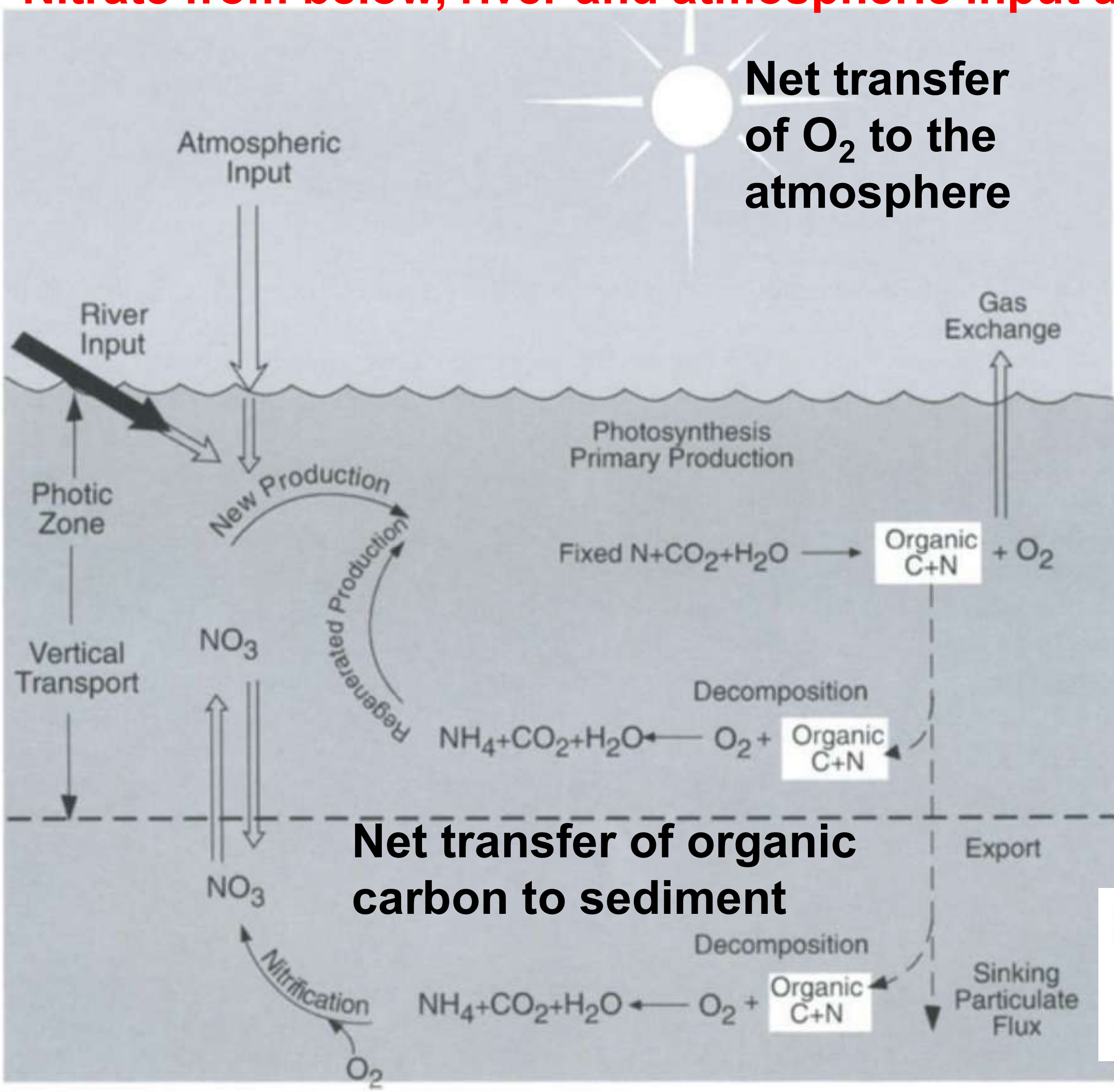
- Most elements in the periodic table are involve in biological cycling. Some are essential nutrients, others are carried along passively and/or form complexes with organic compounds released by organism.
- The production of organic material from inorganic substances in the ocean is described by the RKR equation:

$$106\ CO_2 + 16\ HNO_3 + H_3PO_4, + 122\ H_2O + \text{Light} \rightarrow (CH_2O)_{106}(NH_3)_{16}(H_3PO_4)(\text{Plankton protoplasm}) + 138\ O_2$$

- The mean elemental ratio of C:N:P 106:16:1 by atoms is highly conserved and reflects the average composition of marine phytoplankton and their early degradation products
- This is an organic redox reactions carbon and nitrogen are reduced by water. Only phosphorous does not undergo any change in oxidation state
- The reaction is endothermic. Energy from sunlight is stored in the form of high energy C-C bonds (organic biomass) and O_2.
- Heterotrophic organisms in the ocean are dependent on this biomass and oxygen
- This is not a reversible reaction in the strict sense and there is no equilibrium between products and reactants. The Exothermic reverse reaction respiration occurs both in different parts of the same plankton cells and in heterotrophic organisms
- During respiration nutrients are regenerated. The regeneration ratios of P:N:C:O_2 are about 1:16:117:170 and different from the RKR ratios

Biological processes in the ocean

Primary production is composed of two parts: Primary production in the euphotic zone is called production. New production is based mainly on the upward flux of Nitrate from below, river and atmospheric input and some nitrogen fixation.



The new nitrate produces plankton protoplasm and O_2 as per the RKR equation.

Some of the plant material produced is respired in the euphotic zone due to the combined efforts of zooplankton grazing and bacterial respiration. Oxygen is consumed and organic-N is released as ammonia

$$(CH_2O)_{106}(NH_3)_{16}(H_3PO_4) + 106O_2 \rightarrow$$
$$106CO_2 + 16NH_3 + H_3PO_4 + 106H_2O$$

FIG. 10-13 Schematic representation of biological processes in the marine euphotic zone.

Biological processes in the ocean

The released ammonia is preferentially taken up by photoplankton relative to nitrate and the primary production based on this ammonia is called regenerated production.

The f-ratio is used to describe the relative amounts of new and regenerated production. The f-ratio typically varies between 0.03 and 0.3 in the open ocean but can be >0.8 in the coastal ocean.

$$f = \frac{NO_3\text{uptake}}{NO_3 + NH_4\text{uptake}}$$

We define the new + regenerated production as the gross production (P) and the difference of gross minus regenerated production as net production.

$$f = (P - R) / P$$

Liebig's law of the minimum states that under equal conditions of temperature and light the nutrient available in the smallest quantity relative to the requirement of the plant will limit productivity. Plots of PO_4 vs NO_3 in deep water show a slope of slightly less than 6 and a positive intercept on the PO4 axis. This means when the water is upwelled, NO_3 is the nutrient that will run out first. HCO_3^- in is in large excess and never limiting in the ocean. The nitrogen depletion relative to RKR reflects nitrogen loss due to denitrification which occurs in the oxygen minimum zones.

Biological processes in the ocean

- In the Pacific gyre the situation is a bit more complicated during intense El Niño conditions winds are calmer than usual creating a stagnant photic zone and upwelling is suppressed. This limits PO_4 supply. At the same time nitrogen fixing blue green microbes called Trichodesmium bloom. This can shift the system into a phosphorous limited state.
- Phosphorous input into the ocean across the timescale of millennia is controlled by changes in the chemical weathering rates. In comparison nitrogen can also be fixed by microbes when nitrate input from weathering and the atmosphere are insufficient so P controls variation in primary productivity across geological periods.
- Silicic acid (H_4SiO_4) is a necessary nutrient for diatoms who build their shells from opal ($SiO_2 \cdot nH_2O$). Si:P ratios fixed by diatoms range from 16:1 to 23:1. Variability in diatom new production in the Pacific at times of high nutrient concentrations may be linked to limited silicic acid availability

Biological processes in the ocean

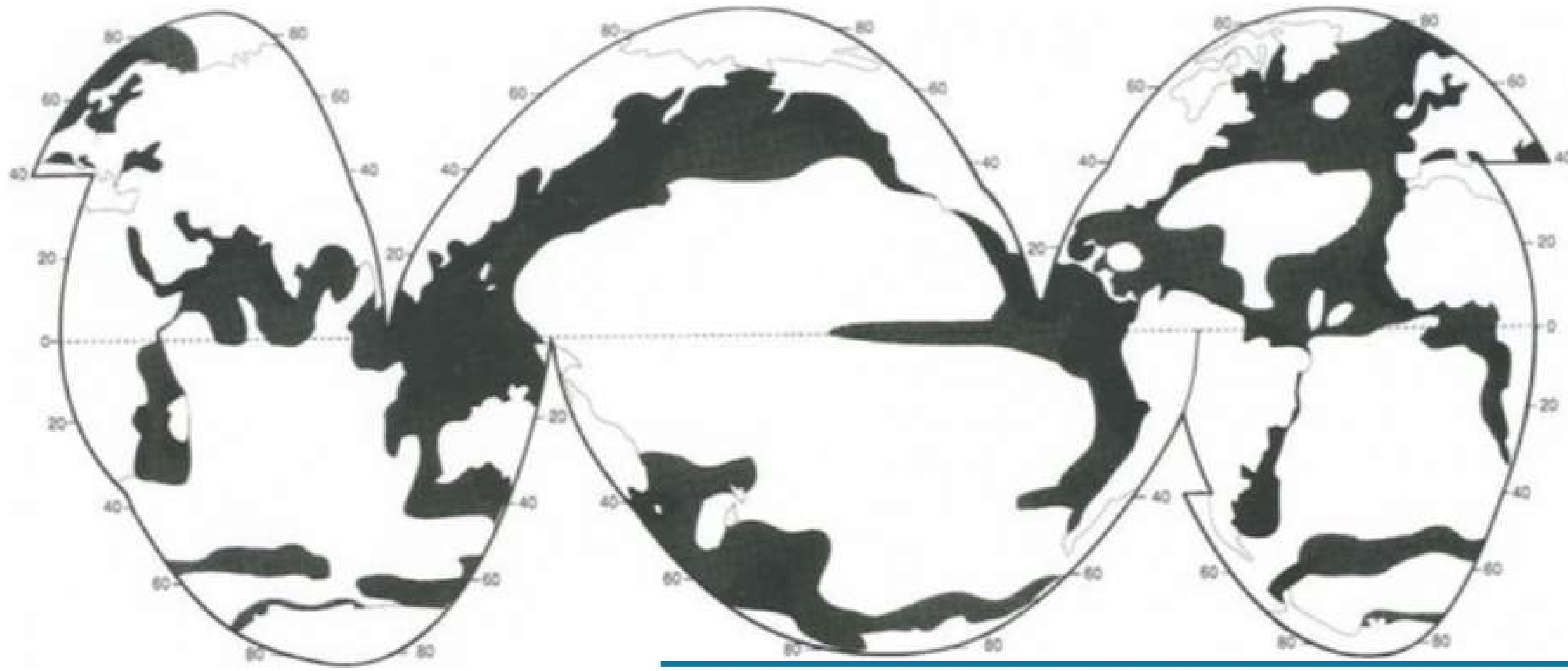


FIG. 10-14 Approximate geographical distibution of primary productivity in the oceans (g C/m^2 per year).

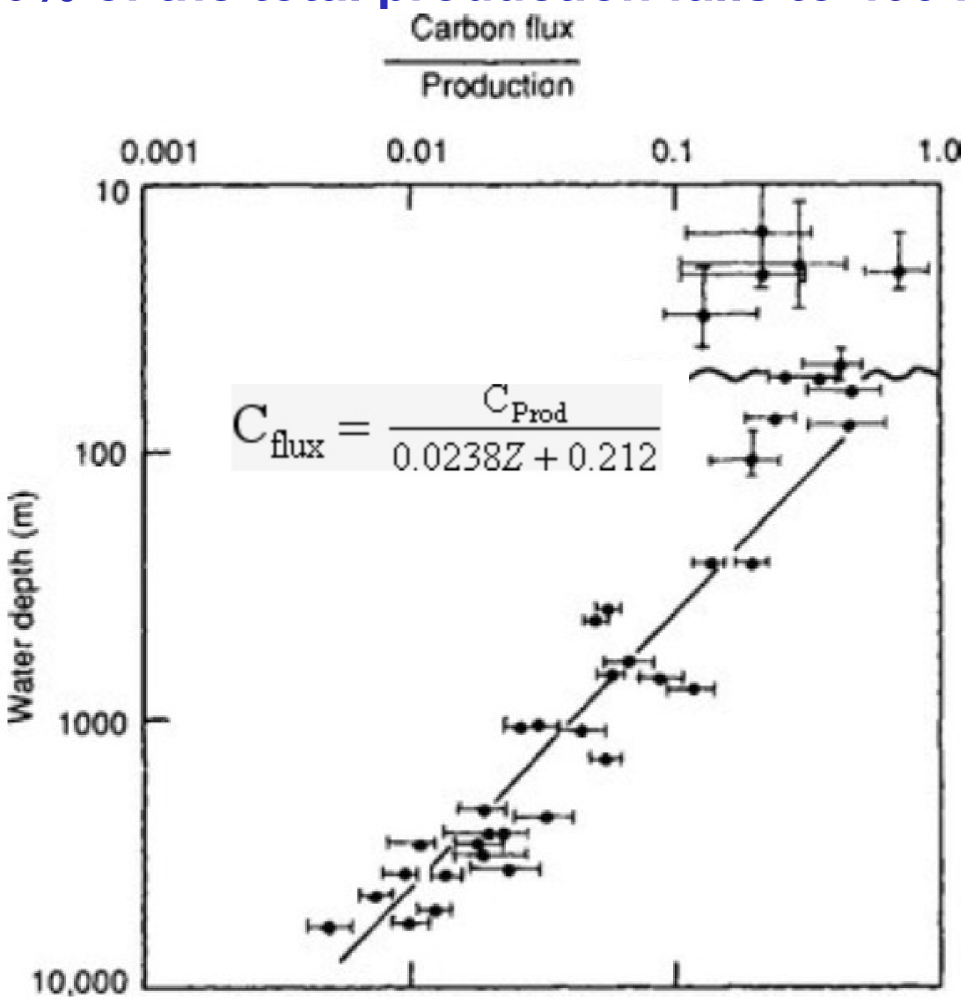
Table 10-5

Distribution of ocean productivity

Province	Percentage of ocean	Area (10^6 km^2)	Mean productivity[a] (g C/m^2/yr)	Total productivity[a] (10^{15} g C/yr)
Open ocean	90	326	50	16.3
Coastal zone[b]	9.9	36	100	3.6
Upwelling areas	0.1	0.36	300	0.1
Total				20

Organic carbon pathways in the ocean

- About 90% of al phytoplankton are eaten by zooplankton and encapsulated into large 0.1-0.3 mm fast sinking (50-500 m/day) fecal pelets
- Sediment trap studies in the ocean have shown that the flux of organic carbon (C_{flux}) at any depth is proportional to the primary productivity (C_{prod}) and inversely proportional to the depth of the water column (z). It can be seen that 10% of the total production falls to 400 m while only 1% reaches 5000 m.



$$C_{flux} = \frac{C_{Prod}}{0.0238Z + 0.212}$$

When depth is transformed to time the slope represents a rate constant for in situ organic carbon loss from the sinking particles. With an average settling rate of 100 m/day the equation becomes:

IG. 10-15 Organic carbon fluxes with epth in the water column normalized to iean annual primary production rates at ie sites of sediment trap deployment. he undulating line indicates the base f the euphotic zone; the horizontal rror bars reflect variations in mean nnual productivity as well as replicate ux measurements during the same eason or over several seasons; vertical rror bars are depth ranges of several ediment trap deployments and uncer-ainities in the exact depth location. (Re-oduced with permission from E. Suess (1980) . articulate organic carbon flux in the oceans – surface oductivity and oxygen utilization, *Nature* **288** : 260–263, Macmillan Magazines.)

$$2.38t = \frac{C_{Prod}}{C_{flux} - 0.212}$$

t=Z/(dZ/dt).

90% of the primary production from falling particles degrades in 4.1 days by the time the particle reaches 400 m depth.

Oceanic reservoirs of organic carbon

1. Most organic carbon in the ocean (97%) is dissolved organic carbon, surface concentrations are 80 µM while below 500 m the average concentration is 40 µM (higher in the Atlantic and lower in the Pacific)
2. The C/N ratio of dissolved organic matter ~17 is greater than the RKR ratio of 6.6. DON has a similar distribution as DOC and seems to be in degradation resistant biomolecules.
3. 95% of the particulate organic matter is detritus (dead).

Table 10-6

Organic carbon reservoirs in the ocean

Depth (m)	Dissolved (10^6 tonnes C)	Approximate concentration (μg C/L)	Particulate (10^6 tonnes C)	Approximate concentration (μg C/L)	Living fraction (10^6 tonnes C)	Approximate % of particulate
0–300[a]	110 000	1000–1500	11 000	100	550[f]	5[f]
300–3800[b]	630 000	500–800	13 000	3–10	<400[g]	<3[g]
Totals	740 000[c] 670 000[d] 1 000 000[e]		24 000[c] 14 000[d] 30 000[e]			

[a]Concentrations vary widely with geographical area and with season. The depth at which the concentration tends to approach a more or less constant level varies widely from 100 to 500 m.

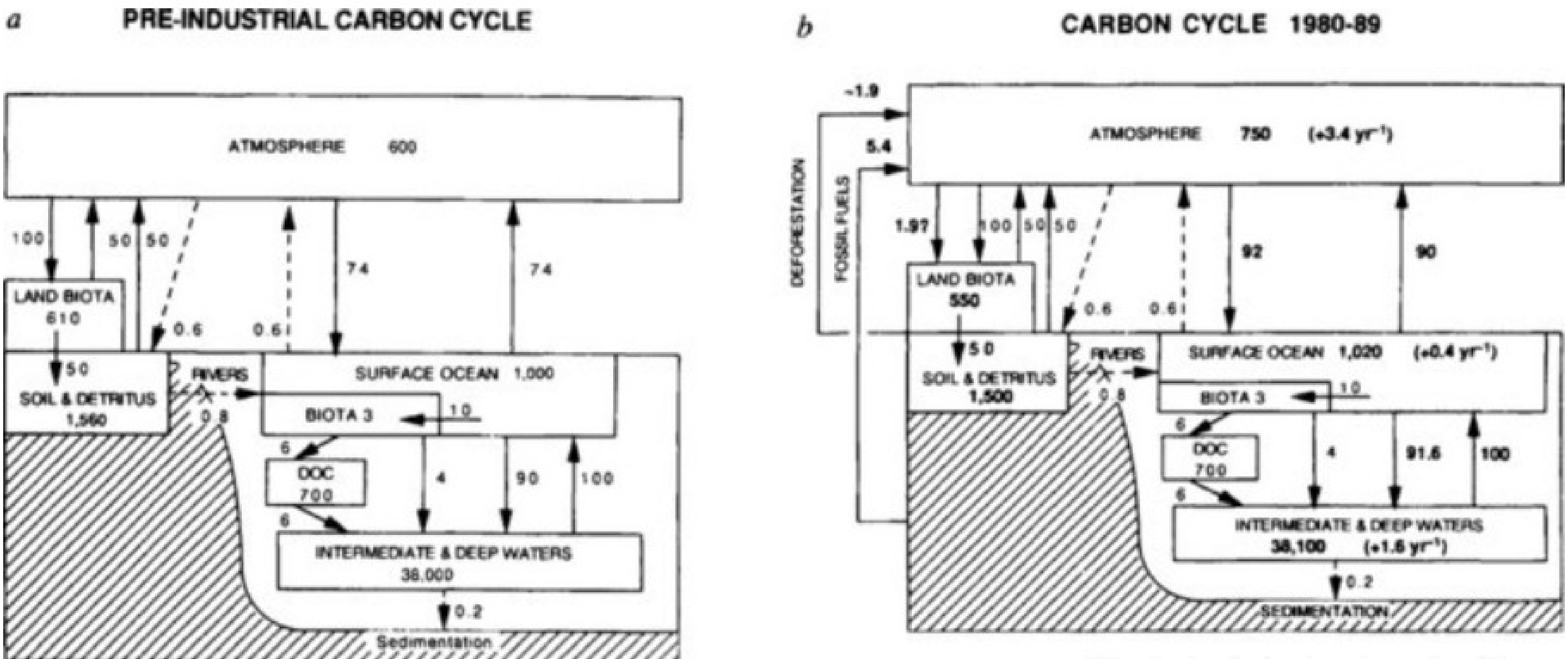
[b]Concentrations more constant.

[c]Williams (1971).

[d]Menzel(1974).

Oceanic budget for organic carbon

Net primary production is 10 Gt C yr^{-1}. Gross primary production is 45 Gt C yr^{-1} 43% of the global total. The average f-ration is ~20%. The turnover rate is fast 1/3 yr^{-1} and 20% to 60% of the exported C is DOC .



The main removal process is burial in sediments. Flux can be calculated as sedimentation rate x weight percent of organic carbon in ocean sediments. The sink by burial (2 x 10^{14} g C yr^{-1}) is much smaller than the production term. If the ocean is in steady state then 98% of the input must be remineralized by respiration and decomposition in the water column or the surface sediments.

FIG. 10-16 A simple schematic of the carbon cycle. Part (a) is the pre-industrial case, and part (b) shows the contemporary reservoirs and fluxes, in Pg C and Pg C/yr, respectively (Pg C = 10^{15} g C). This diagram of the carbon cycle is similar to those presented in Chapters 4 and 11.

(Reprinted by permission from *Nature* (1993). **365:** 119–125, Macmillan.)

Nutrient budgets in the ocean

Surface waters are stripped of nutrients by phytoplankton. However, more than 90% of the primary productivity is released back to the water column as a reverse RKR equation through remineralization and respiration when particles sink. Unlike photosynthesis which is light limited this process can occur across the full water column and in sediments.

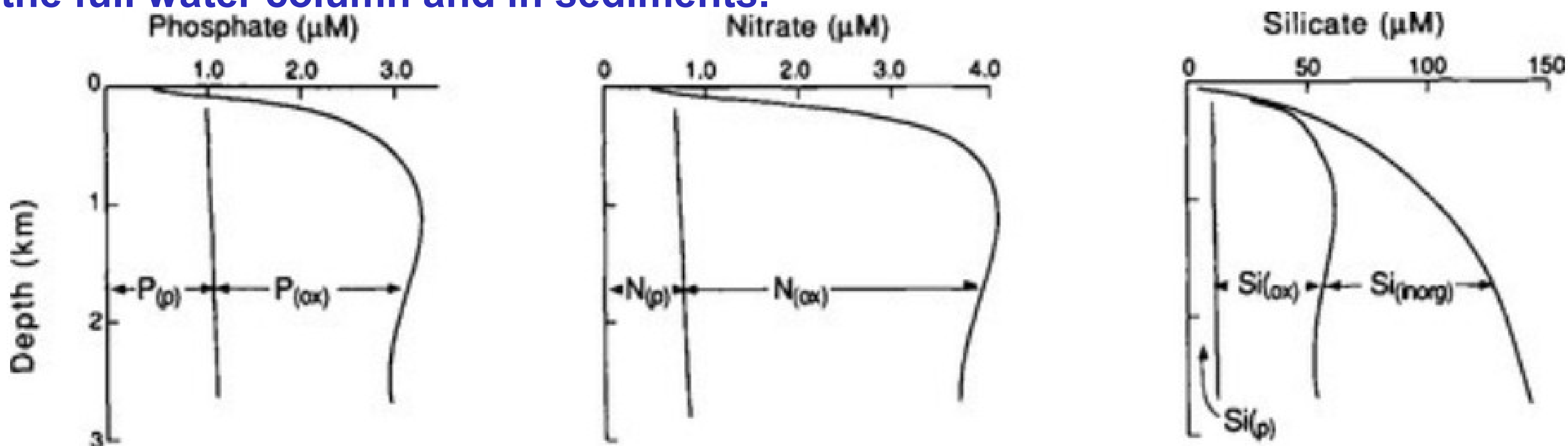


FIG. 10-19 Vertical distributions of various nutrients and their components.

$$P_{total} = P_{preformed} + P_{oxidized} P_{oxidized} = \frac{1}{138} AOU$$

$$N_{total} = N_{preformed} + N_{oxidized} N_{oxidized} = \frac{16}{138} AOU$$

Deep waters contain preformed nutrients that sank with the water and nutrients derived by in situ remineralization of organic particles called oxidative nutrients. Unlike photosynthesis which is light limited this process can occur across the full water column and in sediments.

Remineralization can be estimated from the apparent oxygen utilization (AOU):

$$AOU = O_{2\,sat} - O_{2\,meas}$$

for every mol 138 mol of O_2 consumed in situ on gets 106 mol of CO_2, 16 mol NH_3 and 1 mol H_3PO_4

The carbonate system

The key species required to understand the portioning in the system are alkalinity, H^+, OH^- (pH), pCO_2, H_2CO_3, HCO_3^-, CO_3^{2-} are four equilibrium constants K_H, K_w, K_1 and K_2 connect these.

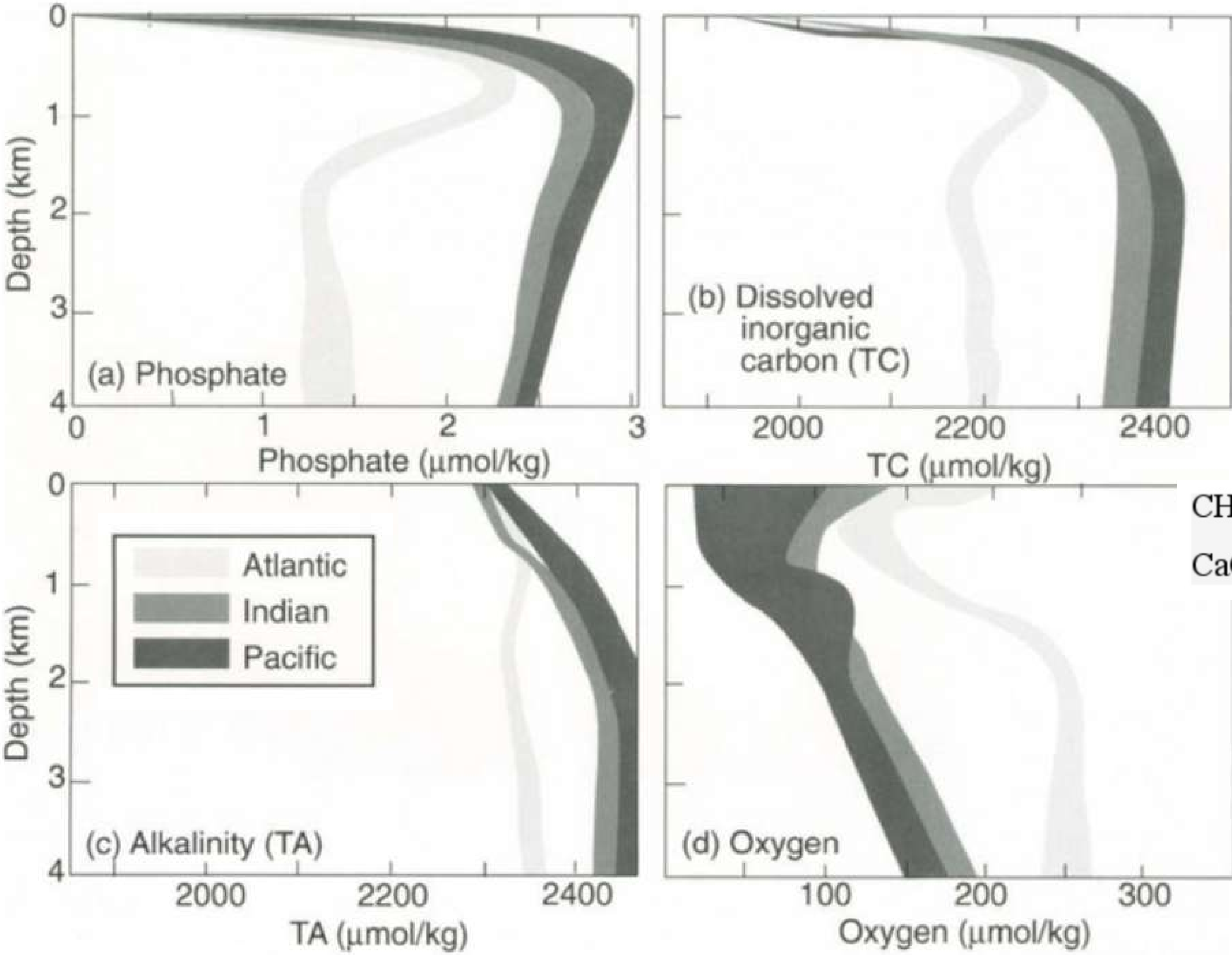


FIG. 10-20 Observed depth profiles of (a) phosphate, (b) dissolved inorganic carbon (TC), (c) alkalinity (TA), and (d) oxygen for the Atlantic, the Indian, and the Pacific Oceans as indicated. Data are

pH profiles and pCO_2 not shown are similar in shape to O_2

$$CH_2O + O_2 = CO_2 + H_2O \quad \Delta(DIC) = 1 \quad \Delta(Alk) = 0$$

$$CaCO_3 = Ca^{2+} + CO_3^{2-} \quad \Delta(DIC) = 1 \quad \Delta(Alk) = 2$$

River input in relation to ocean chemistry

River inputs would produce a very different ocean chemistry

Constituent	Amount in ocean	Amount delivered by rivers to ocean in 10^8 years	Number of times constituents been "renewed" in 10^8 years
SiO_2	0.008	42.6	5300
HCO_3^-	0.19	190.2	1000
Ca^{++}	0.6	48.8	81
K^+	0.5	7.4	15
SO_4^{2-}	3.7	36.7	10
Mg^{++}	1.9	13.3	7
Na^+	14.4	20.7	1.4
Cl^-	26.1	25.4	1
H_2O	1370	3 333 000	2400

Reaction (balanced in terms of mmol of constituent)	Constituents SO_4^{2-}	Ca^{2+}	Cl^-	Na^+	Mg^{2+}	K^+	SiO_2	HCO_3^-		% of total products formed
To be removed from ocean in 10^8 years	382	1220	715	900	554	189	710	3118		
Reaction 1	191	1220	715	900	554	189	710	3500	96 pyrite	3%
									287 kaolinite	8%
Reaction 2	0	1029	715	900	554	189	710	3500	191 "$CaSO_4$"	5%
Reaction 3	0	1029	715	900	502	189	710	3396	52 $MgCO_3$ in magnesium calcite	2%
Reaction 4	0	0	715	900	502	189	710	1338	1029 calcite or aragonite	29%
Reaction 5	0	0	0	185	502	189	710	1338	715 "NaCl"	20%
Reaction 6	0	0	0	185	502	189	639	1338	71 silica	2%
Reaction 7	0	24	0	139	502	189	639	1338	138 sodic montmorillonite	4%
Reaction 8	0	0	0	139	502	189	639	1290	24 calcite or aragonite	1%
Reaction 9	0	0	0	0	502	189	278	1151	417 sodic montmorillonite	12%
Reaction 10	0	0	0	0	0	189	218	147	100 chlorite	3%
Reaction 11	0	0	0	0	0	0	29	−42	378 illite	11%

We take all the ions in the water added to the ocean over 100 million years and then calculate what needs to be removed & find a suitable reaction to make things "right".

11 key precipitation reactions (reverse weathering) are balancing ocean chemistry

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 15th September if you want to get them corrected before the Midsem exam

Numerical

1. Barium has a concentration of 11.7 µg/kg in the ocean and of 628 µg/g in the upper crust. What is it's estimated lifetime in the ocean. Use

$$\log \tau_0 = a \log K_D + b \qquad a\sim1.01;\ b\sim8.4$$

2. CO_2 has a $K_{H\,CO2}$=31 mol $m^{-3}atm^{-1}$ calculate the atmospheric CO_2 residence time with respect to the main loss process (ocean uptake)
3. How how far from the coast will 5 µm sized hematite (5.3 g/cm^3) iron fertilize the Atlantic ocean if the dust plume is originally lifted to 2 km over the Sahara and the trade winds blow it towards South america with 15 m/s wind speed

η_{air}=1.8 x 10^{-5} Pa·s

$$v_{grav} = \frac{\rho_p d^2 C_c}{18\eta\chi_p}$$

4. A) How long will a 0.5 mm foraminifera (2 g/cm^3) take to settle from 50 m depth to the ocean floor at 400 m if the density of ocean water is (1.03 g/cm^3)

η_{ocean}=1.41 x 10^{-3} Pa·s
For 35g/kg 10°C

$$v_{grav} = \frac{(\rho_p - \rho_f)d^2}{18\eta\chi_p}$$

B) Use the intial velocity and $2.38t = \frac{C_{Prod}}{C_{flux} - 0.212}$ to estimate how deep this particle can reach at most. Explain why it can't fall deeper. Also indicate whether fall velocity will actually remain constant or will change with time= depth.

Long answer style

1. Which nutrients typically limit biological productivity in coastal waters and which nutriens are limiting in the middle of the Pacific gyre? Why?

2. Why is a vertical profile showing a strong concentration enhancement at 3000 m depth indicative of a hydrothermal source for the respective element.
3. Why are certain elements depleted in the photic zone?
4. Explain the difference between gross and net ocean productivity?
5. Why can the average ocean composition not be attained by evaporating river water? Which are the top 3 ions (in terms of total surplus amount) that need to be removed and the top 3 minerals precipitated to balance ocean chemicstry?